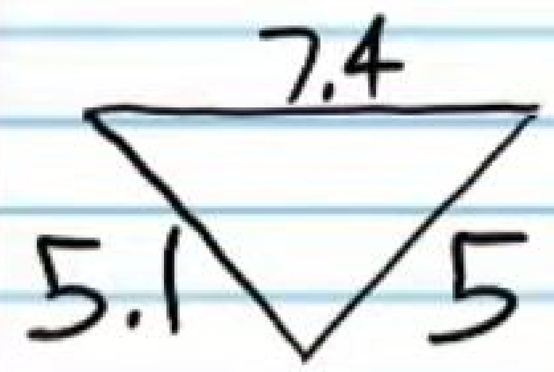
Name
Date
Course

Geometry Test IV

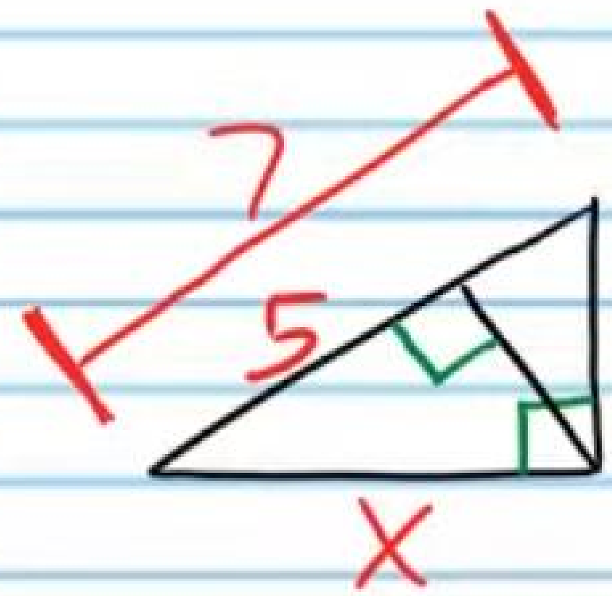
Instructions

I) Write your name, the date, and the course title (i.e. Geometry).
II) Show work when necessary.
III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
IV) Each problem is worth four points for a total of 100 points.
V) A calculator will be necessary for some problems on this test.

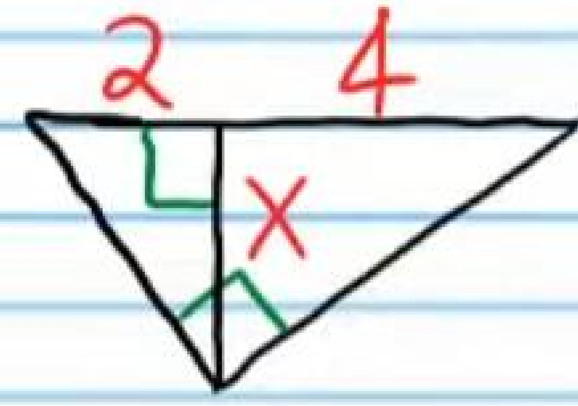
① Is the following triangle an acute, obtuse, or right triangle?



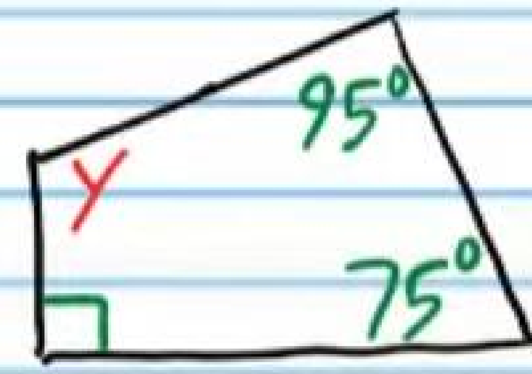
② Find X below.



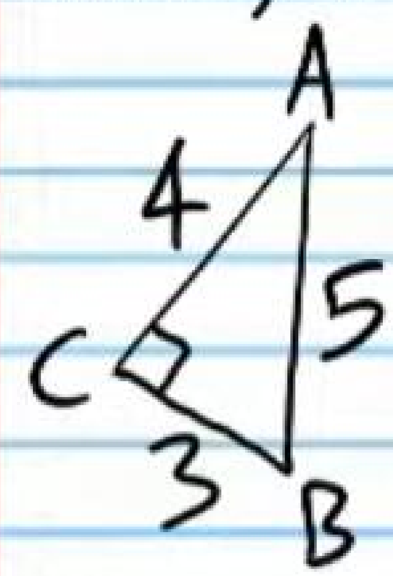
③ Find X.



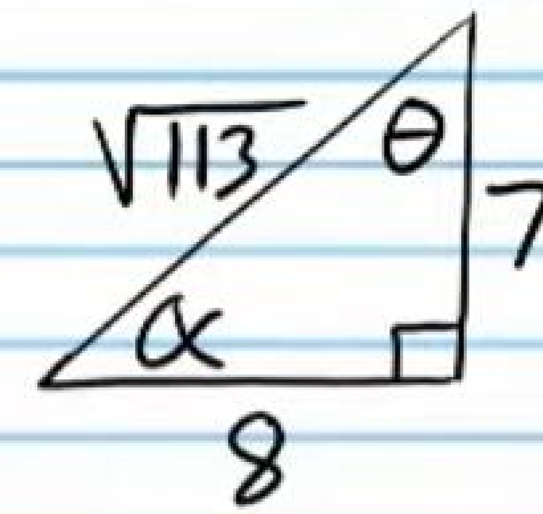
④ Find y.



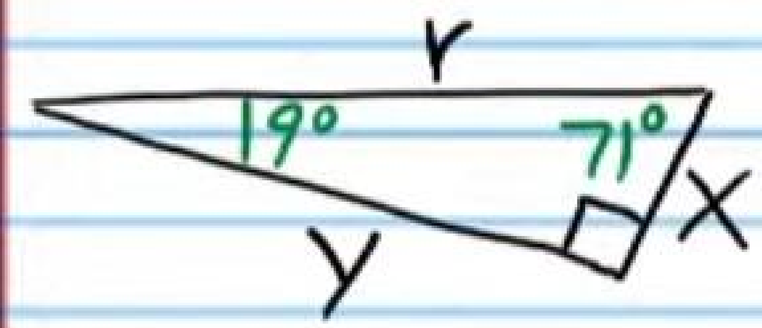
⑤ i) For the triangle below, find Sin B.



ii) For the triangle below, find tan α.



⑥ For the triangle below, find the listed values.
Your answer should have letters. Do not use a calculator.

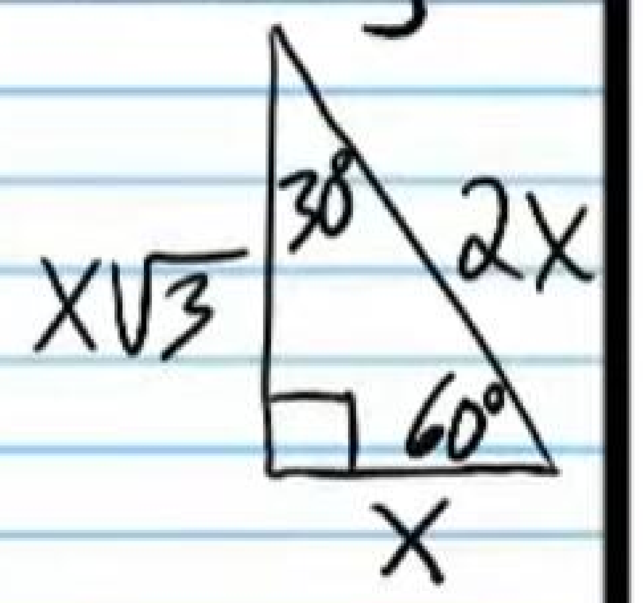


i) $\cos 19°$

ii) $\tan 71°$

⑦ i) Use a calculator to find cos 38°. Round to the nearest thousandth.

Given the triangle below, find the following Values without a calculator.

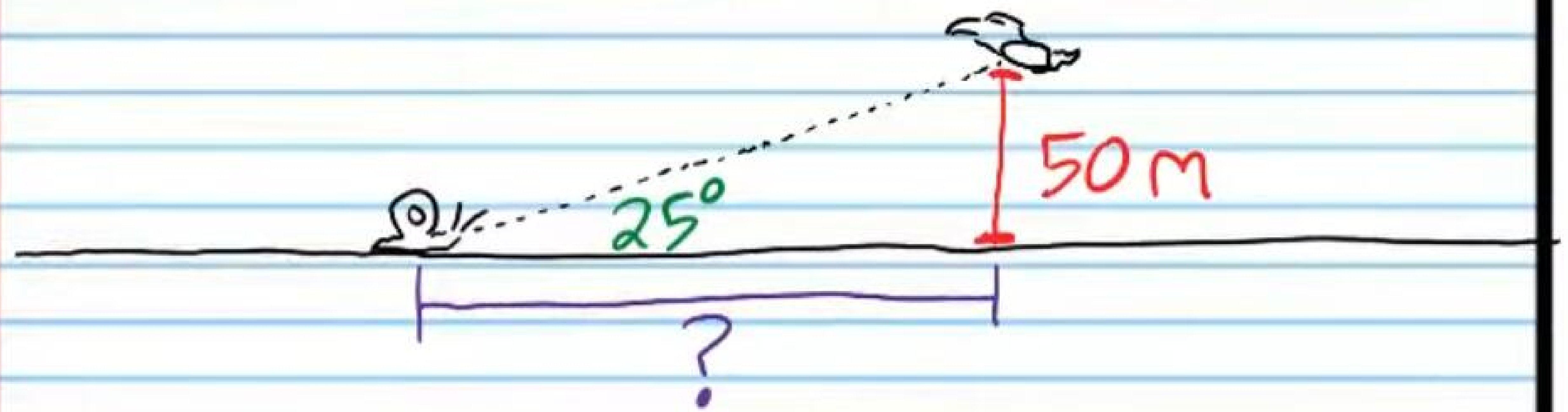


ii) $\sin 30°$

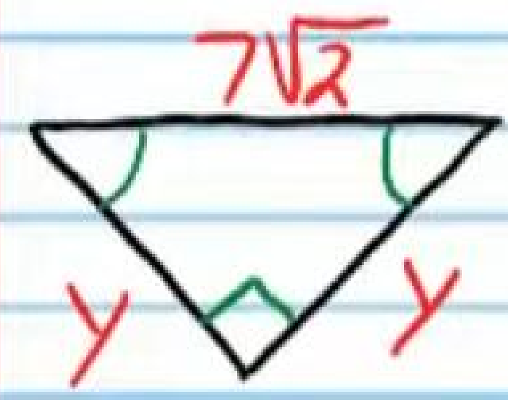
iii) $\cos 30°$

iv) $\tan 60°$

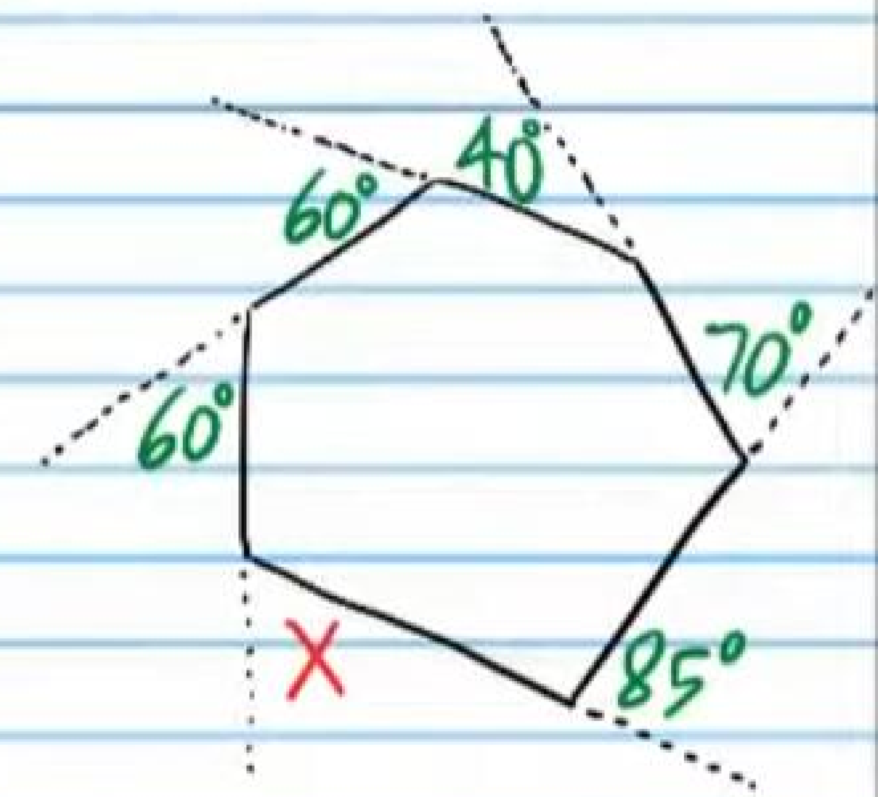
⑧ If a snail looks up at a hawk flying 50 meters above the ground, and the angle of elevation from the snail to the hawk is 25°, find the horizontal distance along the ground between the snail and the hawk. Round to the nearest whole number.



⑨ Find y.



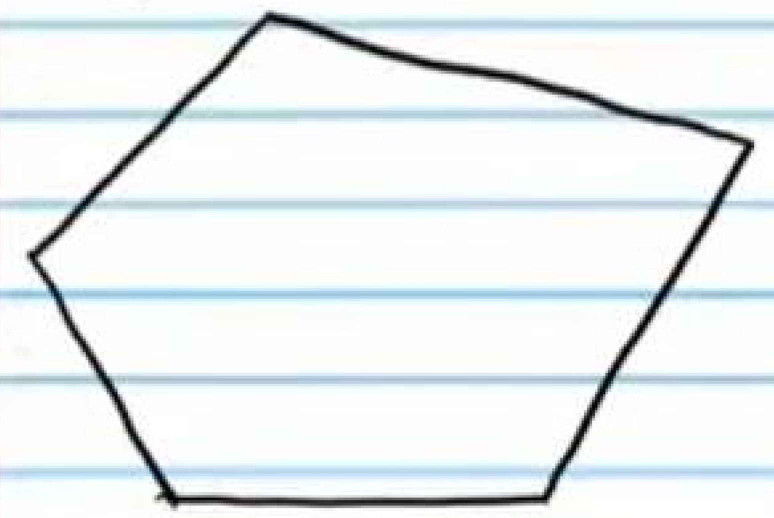
⑩ Find X.



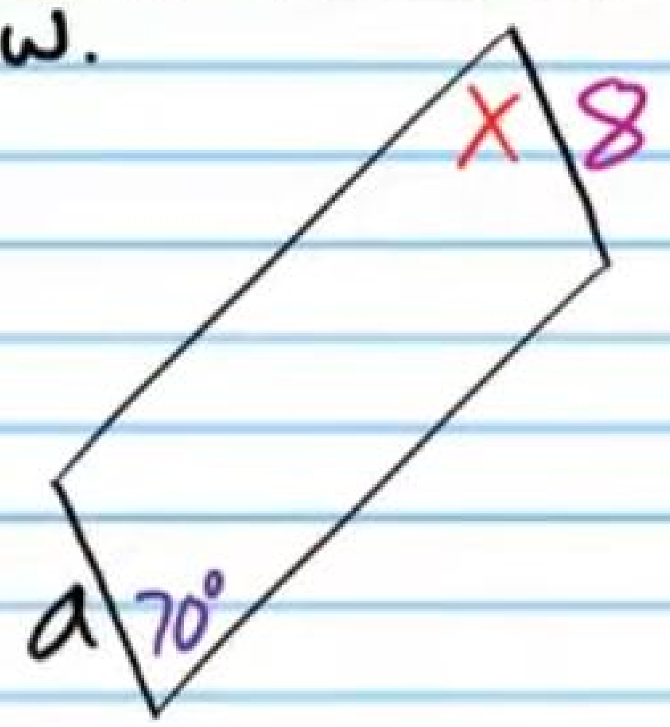
⑪ If the number below is the sum of the measures of the interior angles of a convex polygon, find the number of sides of the polygon.

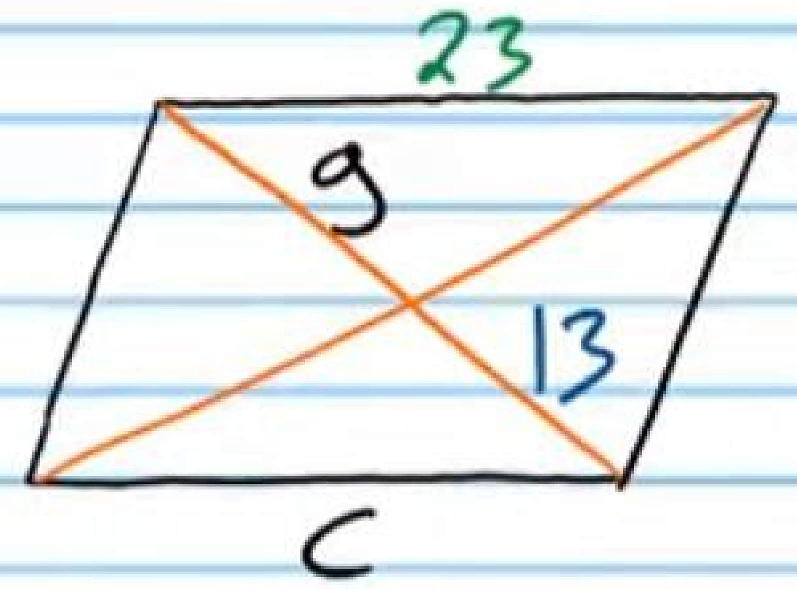
3,780°

⑫ Find the sum of the measures of the interior angles of the following polygon.



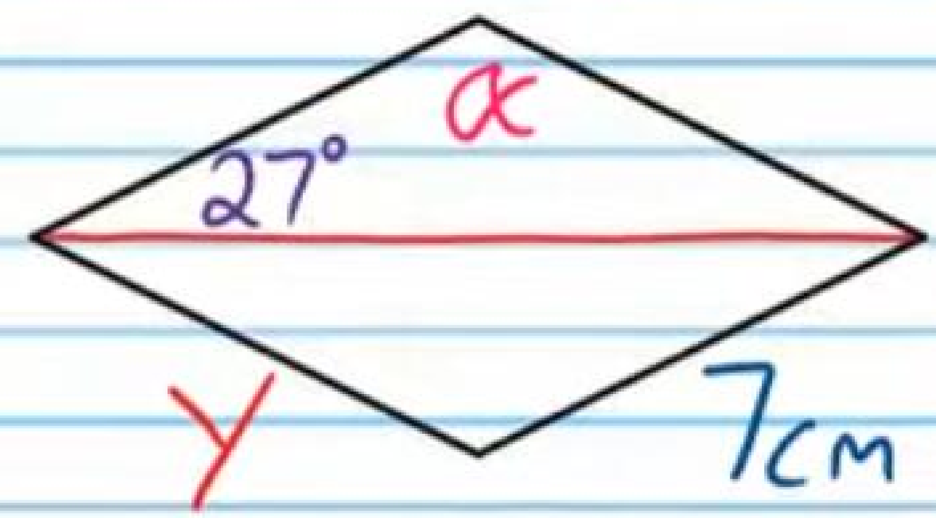
⑬ Find the unknown values of the parallelograms below.

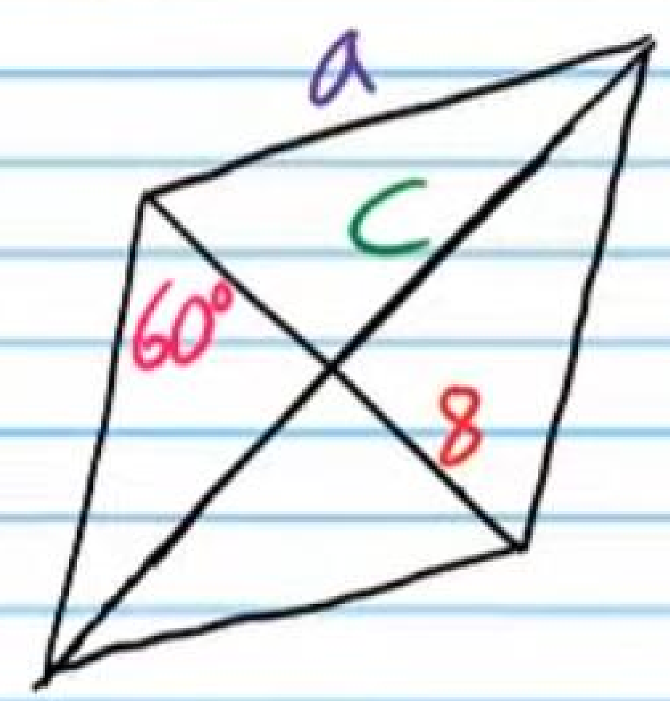




i) a ii) x iii) c iv) g

⑭ Find the unknowns in each rhombus below.





i) y

ii) α

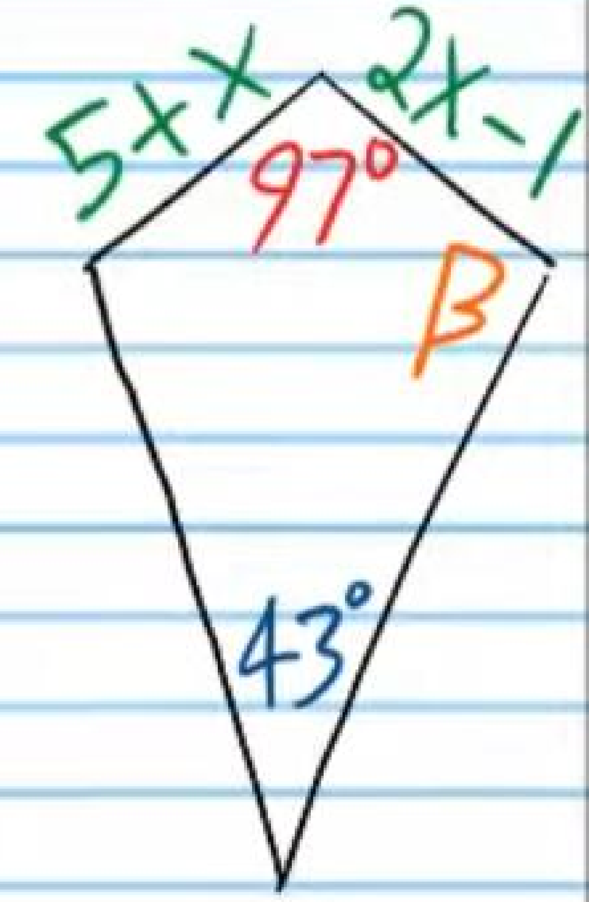
iii) c

iv) a

⑮ Find the unknowns in each kite below.

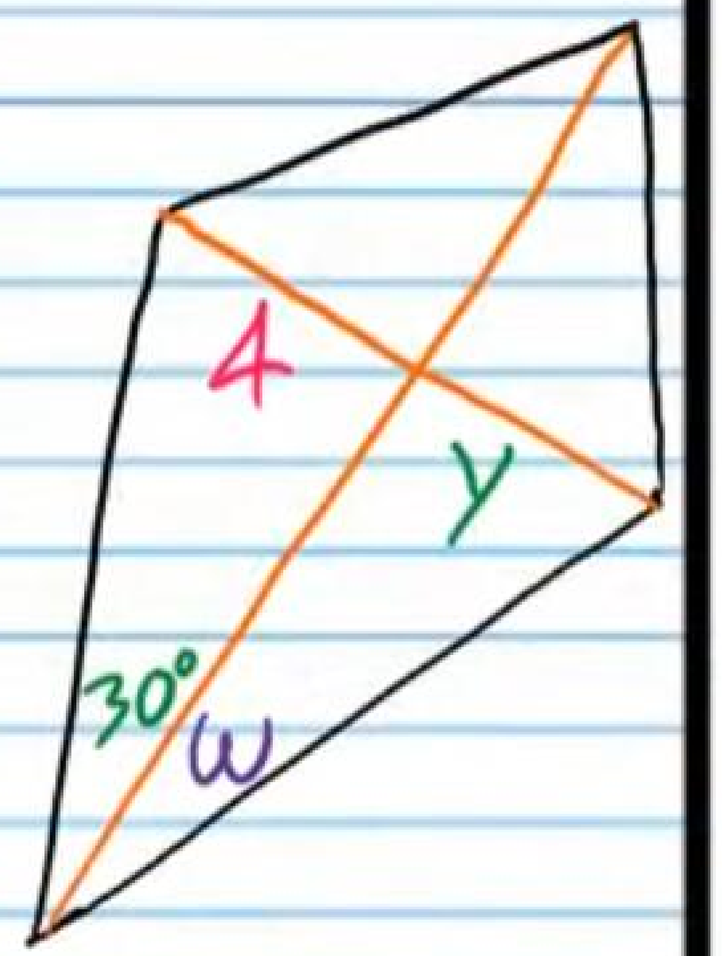
i) β

ii) x



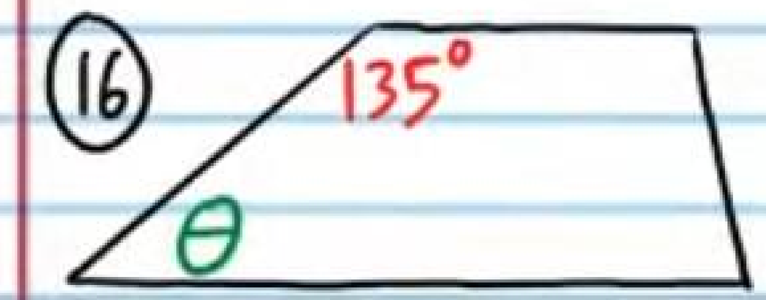
iii) y

iv) w

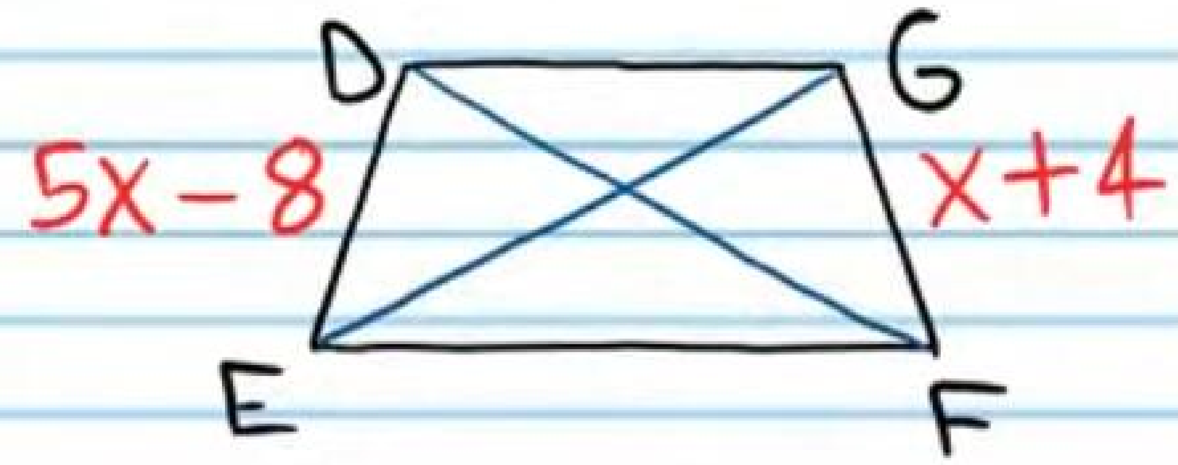


Find the unknowns in each trapezoid below.

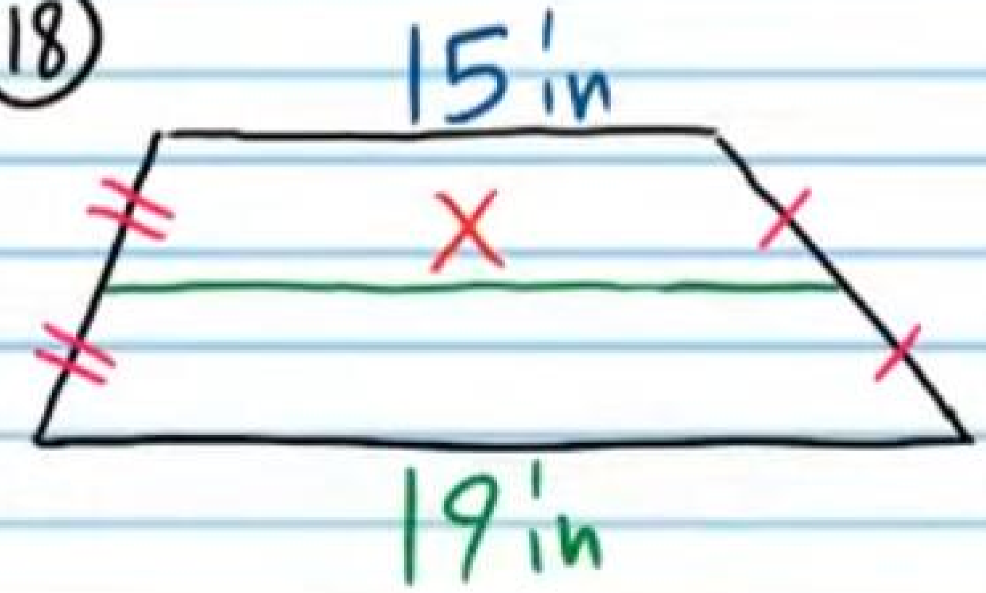
16



17 $\overline{DF} \cong \overline{EG}$

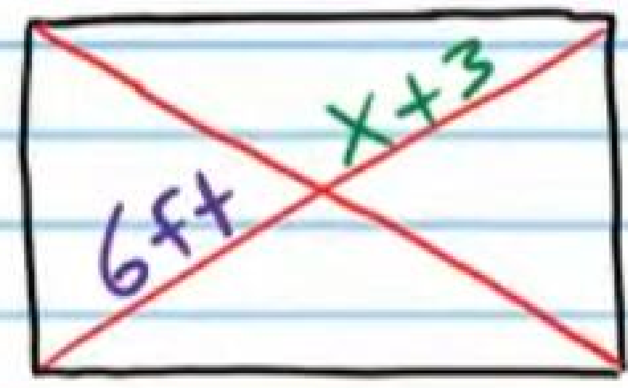


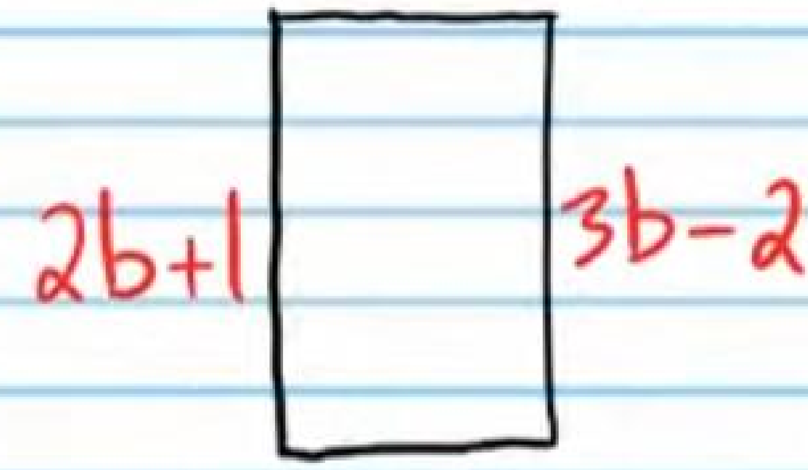
18



Find the unknowns in each rectangle below.

19





i) x

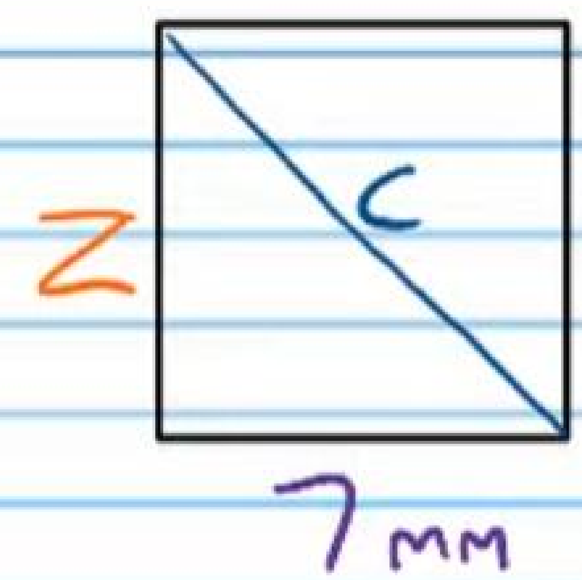
ii) b

20 Find the unknowns in each square below.

i) y

ii) w

iii) Z

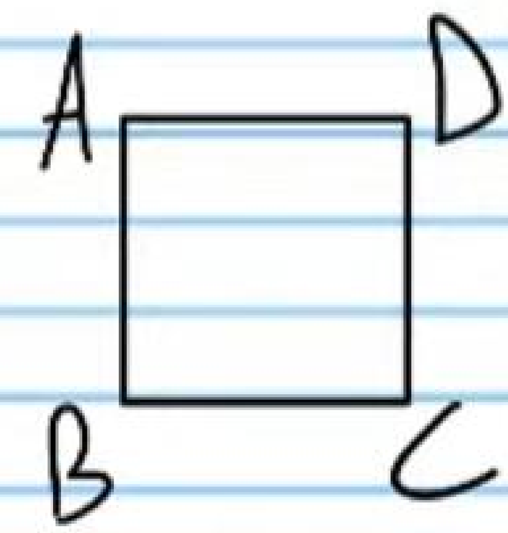


iv) c

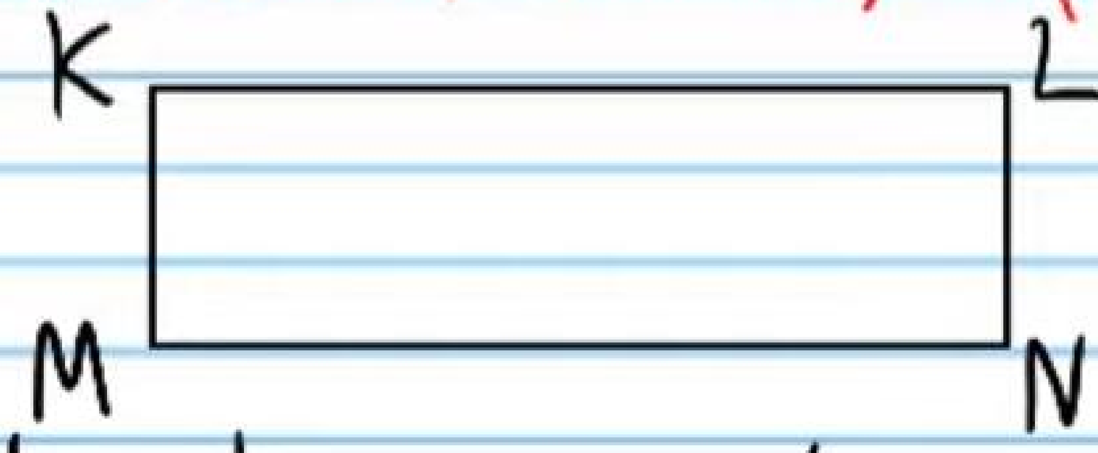
Show that the following statements are true using two-column proofs. You might not use the same number of steps written in Roman numerals. You may use more or less.

(21) If the quadrilateral below is a square, then it is a rhombus.

Statement	Reason
I)	
II)	
III)	



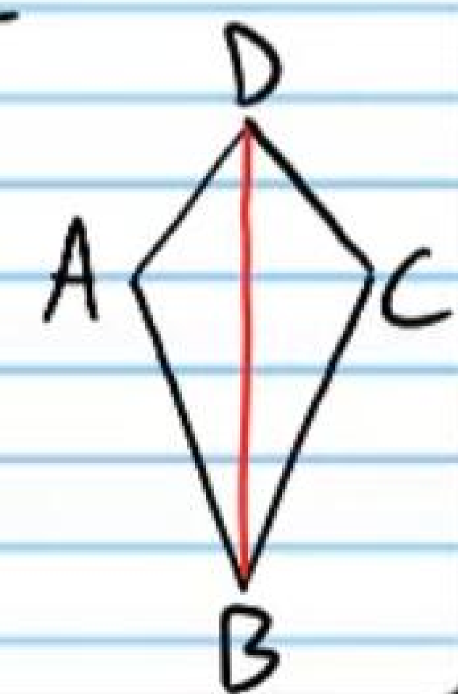
(22) If the quadrilateral below is a rectangle, then its diagonals bisect each other. You may use the Rectangle Parallelogram Theorem. Once you have shown that the rectangle is a parallelogram, then consider the first properties we learned about parallelograms. There are other methods but they require more steps.



Statement	Reason
I)	
II)	
III)	

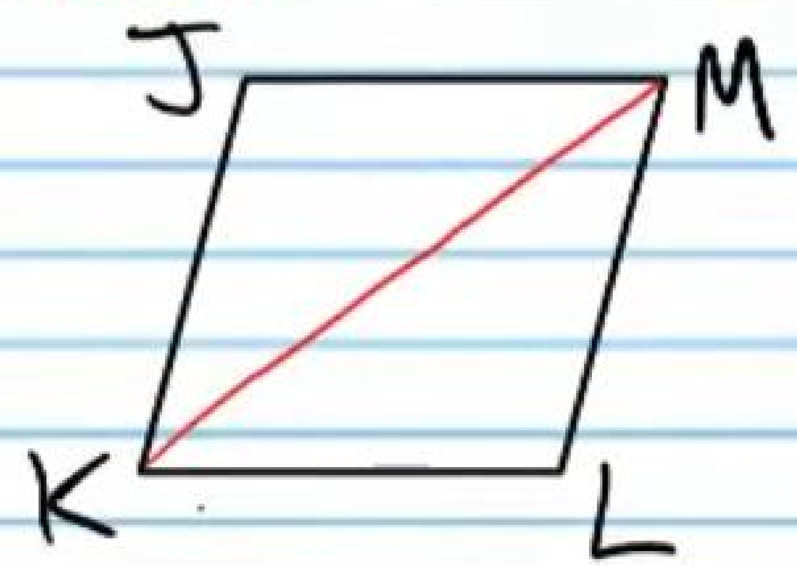
(23) If the quadrilateral below is a kite, then $\angle ADB \cong \angle CDB$.

Statement	Reason
I)	
II)	
III)	



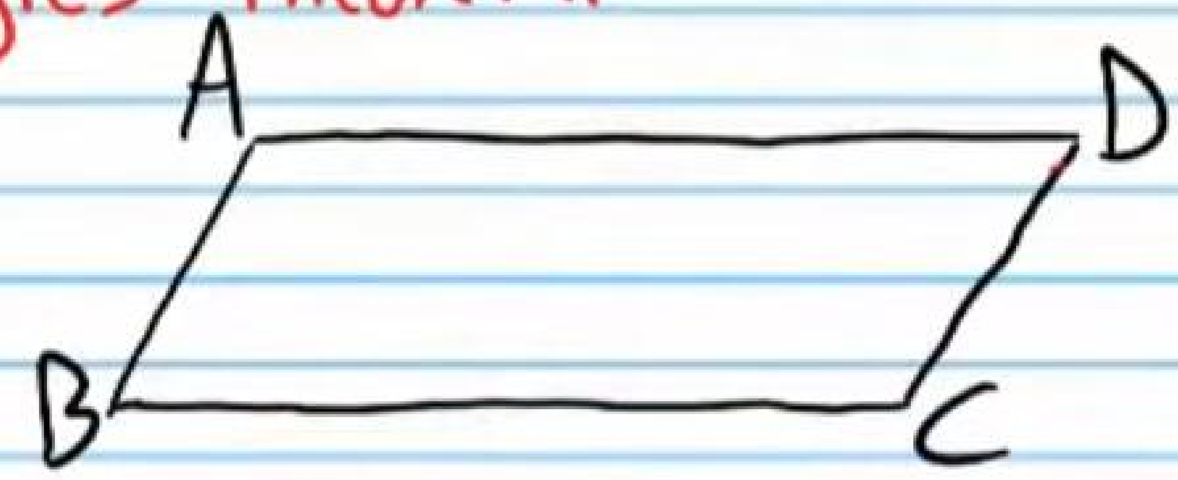
IV)	
V)	

(24) If the quadrilateral below is a rhombus, then it is a parallelogram. You must use the auxiliary line below and the Alternate Interior Angle Converse Theorem.



Statement	Reason
I)	
II)	
III)	
IV)	
V)	
VI)	
VII)	

(25) If the quadrilateral below is a parallelogram, then $\angle A \cong \angle C$ and $\angle B \cong \angle D$. This is the Parallelogram Opposite Angles Theorem.



Statement	Reason
I)	
II)	
III)	
IV)	
V)	
VI)	
VII)	
VIII)	
IX)	
X)	

Geometry Test IV Answers

1. Obtuse
2. $X=\sqrt{35}$
3. $X=2\sqrt{2}$
4. $Y=100^\circ$
5. i) $\sin B = 4/5$
   ii) $\tan \alpha = 7/8$
6. i) y/r
   ii) y/x
7. i) $\cos 38^\circ \approx 0.788$
   ii) $\sin 30^\circ = \frac{1}{2}$
   iii) $\cos 30^\circ = \sqrt{3}/2$
   iv) $\tan 60^\circ = \sqrt{3}$
8. 107 m
9. $Y=7$
10. $X=45^\circ$
11. 23
12. 540°
13. i) $a=8$
    ii) $X=70^\circ$
    iii) $c=23$
    iv) $g=13$
14. i) $y=7$ cm
    ii) $\alpha = 126^\circ$
    iii) $c = 8\sqrt{3}$
    iv) $a=16$
15. i) $\beta = 110^\circ$
    ii) $X=6$
    iii) $y=4$
    iv) $W=30^\circ$
16. $\theta = 45^\circ$
17. $X=3$
18. $X=17$ in
19. i) $X=3$
    ii) $b=3$
20. i) $Y=10\sqrt{2}$ cm
    ii) $W=45^\circ$
    iii) $Z=7$ mm
    iv) $c=7\sqrt{2}$ mm

* Your proofs may be slightly different from those given below, but they could still be correct if they make sense.

21.

Statement	Reason
I) ABCD is a square.	Given.
II) $\overline{AB} \cong \overline{BC} \cong \overline{CD} \cong \overline{AD}$	Def. of a square.
III) ABCD is a rhombus.	Def of a rhombus.

22.

Statement	Reason
I) KMNL is a rectangle.	Given.
II) KMNL is a parallelogram.	▭-▱ Thm.
III) $\overline{KN}$ & $\overline{ML}$ bisect each other.	▱ Diagonals Thm.

(23)

Statement	Reason
I) ABCD is a kite.	Given.
II) $\overline{AD} \cong \overline{DC}$ & $\overline{AB} \cong \overline{BC}$	Def of a kite.
III) $\overline{BD} \cong \overline{BD}$	Ref. Prop of $\cong$.
IV) $\Delta ABD \cong \Delta CBD$	SSS
V) $\angle ADB \cong \angle CDB$	CPCTC

(24)

Statement	Reason
I) JKLM is a rhombus.	Given.
II) $\overline{KJ} \cong \overline{JM}$ & $\overline{KL} \cong \overline{LM}$	Def of a rhombus
III) $\overline{KM} \cong \overline{KM}$	Ref. Prop. of $\cong$.
IV) $\Delta KJM \cong \Delta MLK$	SSS
V) $\angle JKM \cong \angle KML$ & $\angle MKL \cong \angle JMK$	CPCTC.
VI) $\overline{KJ} \parallel \overline{ML}$ & $\overline{JM} \parallel \overline{KL}$	AIACT
VII) JKLM is a parallelogram.	Def. of a $\square$.

(25)

Statement	Reason
I) ABCD is a parallelogram	Given.
II) $\overline{AB} \parallel \overline{DC}$ & $\overline{AD} \parallel \overline{BC}$.	Def. of $\square$
III) $\angle BAC \cong \angle DCA$ & $\angle DAC \cong \angle BCA$	AIAT
IV) $\overline{AC} \cong \overline{AC}$	Ref. Prop of $\cong$
V) $\Delta BAC \cong \Delta DCA$	ASA
VI) $\angle B \cong \angle D$	CPCTC
VII) $\angle DBC \cong \angle ADB$ & $\angle ABD \cong \angle BDC$	AIAT
VIII) $\overline{BD} \cong \overline{BD}$	Ref. Prop of $\cong$
IX) $\Delta ABD \cong \Delta CDB$	ASA
X) $\angle A \cong \angle C$	CPCTC